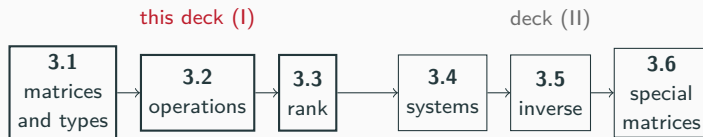


Linear Algebra

3 Matrices (I)

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Chapter map



Central idea: a matrix arranges numbers in rows and columns; its columns are vectors (Ch. 2), and with it the computations for systems are done all at once.

Exercises: Sheet 3, ex. 39–65 (original numbering).

3.1 Matrices and types

Definition

An $m \times n$ **matrix** over $\mathbb{K}$ is an array of mn elements of $\mathbb{K}$ in m rows and n columns:

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad a_{ij} : \text{ row } i, \text{ column } j.$$

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Example (this course's running matrix). The columns are the vectors $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ from Ch. 2:

$$C = \begin{bmatrix} 2 & 0 & 0 \\ -6 & 2 & -3 \\ -6 & 0 & -1 \end{bmatrix} = [\mathbf{c}_1 \quad \mathbf{c}_2 \quad \mathbf{c}_3], \quad c_{21} = -6, \quad c_{23} = -3.$$

Types of matrices

- **row** $1 \times n$ / **column** $m \times 1$
- **zero** $0_{m \times n}$: all elements 0
- **square** of order n : $n \times n$;
main diagonal $a_{11}, a_{22}, \dots, a_{nn}$

For square matrices:

- **upper triangular**: $a_{ij} = 0$ if $i > j$
- **lower triangular**: $a_{ij} = 0$ if $i < j$
- **diagonal**: $a_{ij} = 0$ if $i \neq j$
- **identity** I_n : diagonal with $a_{ii} = 1$

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$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} \text{ upper triangular,}$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \text{diag}(2, -1, 3), \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

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Matrices form a vector space

With entrywise addition and scalar multiplication (3.2), the $m \times n$ matrices over $\mathbb{K}$ form a **vector space** (Ch. 2).

- Standard basis: the mn matrices with a single element equal to 1 and all the others 0.
- Dimension: mn . For example, the 2×3 matrices form a space of dimension 6.
- It makes sense to speak of subspaces of matrices, linear combinations and independence.

This is what is done in Ex. 41.

- **Ex. 39** size and elements of a matrix.
- **Ex. 40** recognising the special types.
- **Ex. 41** a subspace of 2×3 matrices and its dimension.

3.2 Operations with matrices

Addition and scalar multiplication

For $A = [a_{ij}]$ and $B = [b_{ij}]$ of the *same size* $m \times n$ and $\alpha \in \mathbb{K}$:

$$A + B = [a_{ij} + b_{ij}], \quad \alpha A = [\alpha a_{ij}].$$

Example.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + 2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 5 & 4 \end{bmatrix}.$$

Matrices of different sizes cannot be added.

Matrix product

Definition

If A is $m \times n$ and B is $n \times p$, the product AB is $m \times p$, with

$$(AB)_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{in}b_{nk} = \sum_{j=1}^n a_{ij}b_{jk} \quad (\text{row } i \text{ of } A \text{ times column } k \text{ of } B).$$

It exists only if the no. of columns of A equals the no. of rows of B : $(m \times \underline{n})(\underline{n} \times p) = m \times p$.

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It exists only if the no. of columns of A equals the no. of rows of B : $(m \times n)(n \times p) = m \times p$.

It is not commutative:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}.$$

Even when AB and BA both exist, in general $AB \neq BA$. If $AB = BA$, A and B are said to *commute*.

Matrix times vector: a linear combination

If $A = [\mathbf{a}_1 \cdots \mathbf{a}_n]$ (columns) and $\mathbf{x} = (x_1, \dots, x_n)$ as a column:

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

$A\mathbf{x}$ is a linear combination of the columns of A , with coefficients $x_1, \dots, x_n$.

Matrix times vector: a linear combination

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$A\mathbf{x}$ is a linear combination of the columns of A , with coefficients $x_1, \dots, x_n$.

Example.

$$C \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ -6 \\ -6 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ -11 \\ -9 \end{bmatrix}.$$

This is the bridge to Ch. 2: asking whether $\mathbf{b}$ is a linear combination of the columns of A is asking whether $A\mathbf{x} = \mathbf{b}$ has a solution (3.4).

Properties of the product

Whenever the operations are defined:

- $(AB)C = A(BC)$
- $A(B + C) = AB + AC$
- $(A + B)C = AC + BC$
- $\alpha(AB) = (\alpha A)B = A(\alpha B)$
- $I_m A = A = A I_n$ (A $m \times n$)
- powers: $A^2 = AA$, $A^0 = I$

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- powers: $A^2 = AA$, $A^0 = I$

Caution: rules that do *not* hold

- $AB \neq BA$ in general, so $(A + B)^2 = A^2 + AB + BA + B^2$ (not $A^2 + 2AB + B^2$).
- $AB = 0$ does not force $A = 0$ or $B = 0$: $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Definitions

- **Transpose** of $A = [a_{ij}]$ ($m \times n$): $A^T = [a_{ji}]$ ($n \times m$) — rows become columns.
- **Trace** of a square A : $\text{tr } A = a_{11} + a_{22} + \cdots + a_{nn}$.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}, \quad \text{tr } C = 2 + 2 + (-1) = 3.$$

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Properties: $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(\alpha A)^T = \alpha A^T$, $(AB)^T = B^T A^T$ (the order is reversed).

The rotation of the plane through the angle θ (Ch. 1) is the matrix

$$R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} : \quad R_\theta \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \cos \theta - y \sin \theta \\ x \sin \theta + y \cos \theta \end{bmatrix}.$$

LINEAR ALGEBRA IN PRACTICE: composing rotations

The rotation of the plane through the angle θ (Ch. 1) is the matrix

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Rotating by β and then by α is multiplying the matrices:

$$R_\alpha R_\beta = R_{\alpha+\beta} \quad (\text{the formulae for } \cos(\alpha + \beta) \text{ and } \sin(\alpha + \beta)).$$

Composing transformations = multiplying matrices. This is how graphics software chains rotations, scalings and projections.

Compare with Ch. 1: multiplying by $\text{cis } \alpha$ and then by $\text{cis } \beta$ gives $\text{cis}(\alpha + \beta)$.

- **Ex. 42** products and powers. Always check the sizes before multiplying.
- **Ex. 43** operations and simple equations; in d), the matrices that commute with A .
- **Ex. 44** which products are possible; traces.

Before multiplying

Check the sizes. If a product does not exist, say so and justify it (for example, ' 2×1 by 2×1 : not defined'). There are such products, on purpose, in Ex. 43 and 44.

3.3 Rank

The rows of an $m \times n$ matrix are vectors in $\mathbb{K}^n$; the columns are vectors in $\mathbb{K}^m$.

Theorem

In a matrix, the maximum number of lin. indep. *rows* equals the maximum number of lin. indep. *columns*.

Definition

This number is the **rank** of A , car A .

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Theorem

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Definition

This number is the **rank** of A , $\text{car } A$.

$$\text{car} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 5 \end{bmatrix} = 2, \quad \text{car } 0_{m \times n} = 0, \quad \text{car } I_n = n, \quad \text{car } A \leq \min(m, n).$$

Theorem

The rank *does not change* under **elementary operations** on rows (or on columns):

- | | |
|---------------------------------------|------------------------------------|
| 1. swap two rows | $L_i \leftrightarrow L_j$ |
| 2. multiply a row by $\alpha \neq 0$ | $L_i \rightarrow \alpha L_i$ |
| 3. add to a row a multiple of another | $L_i \rightarrow L_i + \alpha L_j$ |

These are Theorems IV and V of Ch. 2 (plus the swap, which does not change the set of vectors).

Elementary operations

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These are Theorems IV and V of Ch. 2 (plus the swap, which does not change the set of vectors).

Row reduction: use elementary operations to bring the matrix to **echelon form** (zeros below each pivot). **car A = no. of non-zero rows of the echelon form.**

Row reduction examples

$$C = \begin{bmatrix} 2 & 0 & 0 \\ -6 & 2 & -3 \\ -6 & 0 & -1 \end{bmatrix} \xrightarrow[\substack{L_2 \rightarrow L_2 + 3L_1 \\ L_3 \rightarrow L_3 + 3L_1}]{} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & -3 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{car } C = 3.$$

Row reduction examples

$$C = \begin{bmatrix} 2 & 0 & 0 \\ -6 & 2 & -3 \\ -6 & 0 & -1 \end{bmatrix} \xrightarrow[L_3 \rightarrow L_3 + 3L_1]{L_2 \rightarrow L_2 + 3L_1} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & -3 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{car } C = 3.$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow[L_3 \rightarrow L_3 - 7L_1]{L_2 \rightarrow L_2 - 4L_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix} \xrightarrow{L_3 \rightarrow L_3 - 2L_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{car} = 2.$$

In Ch. 2 we saw that $(7, 8, 9) = 2(4, 5, 6) - (1, 2, 3)$: the zero row of the echelon form is that dependence.

Example. $A_k = \begin{bmatrix} 1 & 1 \\ 2 & k \end{bmatrix} : \xrightarrow{L_2 \rightarrow L_2 - 2L_1} \begin{bmatrix} 1 & 1 \\ 0 & k - 2 \end{bmatrix}.$

$$\text{car } A_k = \begin{cases} 2, & k \neq 2 \\ 1, & k = 2. \end{cases}$$

Rank with a parameter

Example. $A_k = \begin{bmatrix} 1 & 1 \\ 2 & k \end{bmatrix} : \xrightarrow{L_2 \rightarrow L_2 - 2L_1} \begin{bmatrix} 1 & 1 \\ 0 & k - 2 \end{bmatrix}.$

$$\text{car } A_k = \begin{cases} 2, & k \neq 2 \\ 1, & k = 2. \end{cases}$$

Definition

A square matrix A of order n with $\text{car } A = n$ is called **non-singular**; otherwise, **singular**.

In 3.5 we shall see that 'non-singular' is the same as *invertible*.

- **Ex. 45** rank by row reduction.
- **Ex. 46–47** rank as a function of a parameter k ; values of k for which the matrix is non-singular.

KEY TAKEAWAYS — Matrices (I)

1. A matrix arranges numbers in rows and columns; the columns are vectors, and the matrices of the same size form a vector space.
2. The matrix product is computed row by column, exists only for compatible sizes and, in general, is not commutative.
3. Multiplying a matrix by a vector means forming a linear combination of its columns.
4. The rank is the maximum number of lin. indep. rows (or columns); elementary operations do not change it.
5. To compute it, bring the matrix to echelon form and count the non-zero rows.

Next (Matrices II): systems, the inverse and special matrices.

References (full list: course bibliography)

- Santana, A. P., Queiró, J. F., *Introdução à Álgebra Linear*, Gradiva (in Portuguese).
- Lipschutz, S., Lipson, M., *Schaum's Outline of Linear Algebra*, McGraw-Hill — many solved exercises.
- Strang, G., *Introduction to Linear Algebra*, Wellesley-Cambridge Press.

Resources — Sheet 3 (exercises 39–65) and its solutions.